

Suppose $z = f(x, y)$ is a real-valued bivariate function. If we ‘freeze’ either x or y , then we yield a ‘slice’ of f ’s surface.

Equivalently, given a point (x_0, y_0) in D_f , we can let *one* of these variables vary to yield the same slices. Notice, however, that sometimes, even if we start with the fixed point (x_0, y_0) , the x -slices and the y -slices can look very different. Specifically, we try to ascertain the tangent lines.

The slopes of these tangent lines are different. Specifically, however, the slopes of these tangent lines are essentially single-variable derivatives where either the x - or y - coordinate is held fixed at x_0 or y_0 . These objects are known as **partial derivatives**:

Definition. Let $z = f(x, y)$ be a real-valued bivariate function. The **partial derivative of f with respect to x** is the function:

$$\frac{\partial}{\partial x} \left[f(x, y) \right] = \frac{\partial f}{\partial x}(x, y) = f_x(x, y) = \lim_{h \rightarrow 0} \frac{f(x + h, y) - f(x, y)}{h},$$

provided that the limit exists.

Example. Consider the function $f(x, y) = x^2 + 3xy + y - 1$. Compute $f_x(4, -5)$ and $f_y(4, -5)$.

Example. Consider the function $f(x, y) = \frac{2y}{y + \cos(x)}$. Compute $f_x(x, y)$ and $f_y(x, y)$.

Example. Consider the equation $yz - \ln(z) = x + y$. If this equation defines z as a function of x and y , then find $\frac{\partial z}{\partial x}$

Example. The plane $x = 1$ intersects the paraboloid $f(x, y) = x^2 + y^2$ in a parabola. Find the slope of the tangent to the parabola at $P(1, 2, 5)$.

Higher Order Partial Derivatives

Just as in single-variable calculus, we can compute partial derivatives to higher orders as well. In this case, however, we have *lots* of options with how to compute them. For example, if we want to partially differentiate a function $f(x, y)$ twice, we have the following options:

$$(i) \quad f_{xx} = \frac{\partial^2 f}{\partial x^2}$$

$$(ii) \quad f_{yy} = \frac{\partial^2 f}{\partial y^2}$$

$$(iii) \quad f_{xy} = (f_x)_y = \frac{\partial^2 f}{\partial y \partial x}$$

$$(iv) \quad f_{yx} = (f_y)_x = \frac{\partial^2 f}{\partial x \partial y}$$

Example. If $f(x, y) = x \cos(y) + ye^x$, find all of the second-order partial derivatives.

Theorem [Clairaut's Theorem¹] If $f(x, y)$ and its partial derivatives f_x, f_y, f_{xy} , and f_{yx} are defined throughout an open region containing a point (a, b) and are all continuous at (a, b) , then

$$f_{xy}(a, b) = f_{yx}(a, b).$$

Example. Given $f(x, y, z) = x^2 + 5xy + \sin(x) + 7e^x$, verify that $f_{xy} = f_{yx}$.

Example. Given $w = xy + \frac{e^y}{y^2 + 1}$, compute $\frac{\partial^2 w}{\partial x \partial y}$.

NOTE: The continuity conditions of Clairaut's Theorem are necessary. This was famously observed Peano when he considered this function:

$$f(x, y) = \begin{cases} \frac{xy(x^2 - y^2)}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}.$$

It can be shown that f, f_x , and f_y are all continuous everywhere, however f_{xy} and f_{yx} are not continuous at $(0, 0)$. It can then be shown that $f_{xy}(0, 0) = -1 \neq 1 = f_{yx}(0, 0)$ so that $f_{xy} \neq f_{yx}$ and the conclusion of Clairaut's Theorem fails.

¹Clairaut's Theorem applies for higher-order partial derivatives as well—as long as everything is continuous, then it doesn't matter what order to take the partials.